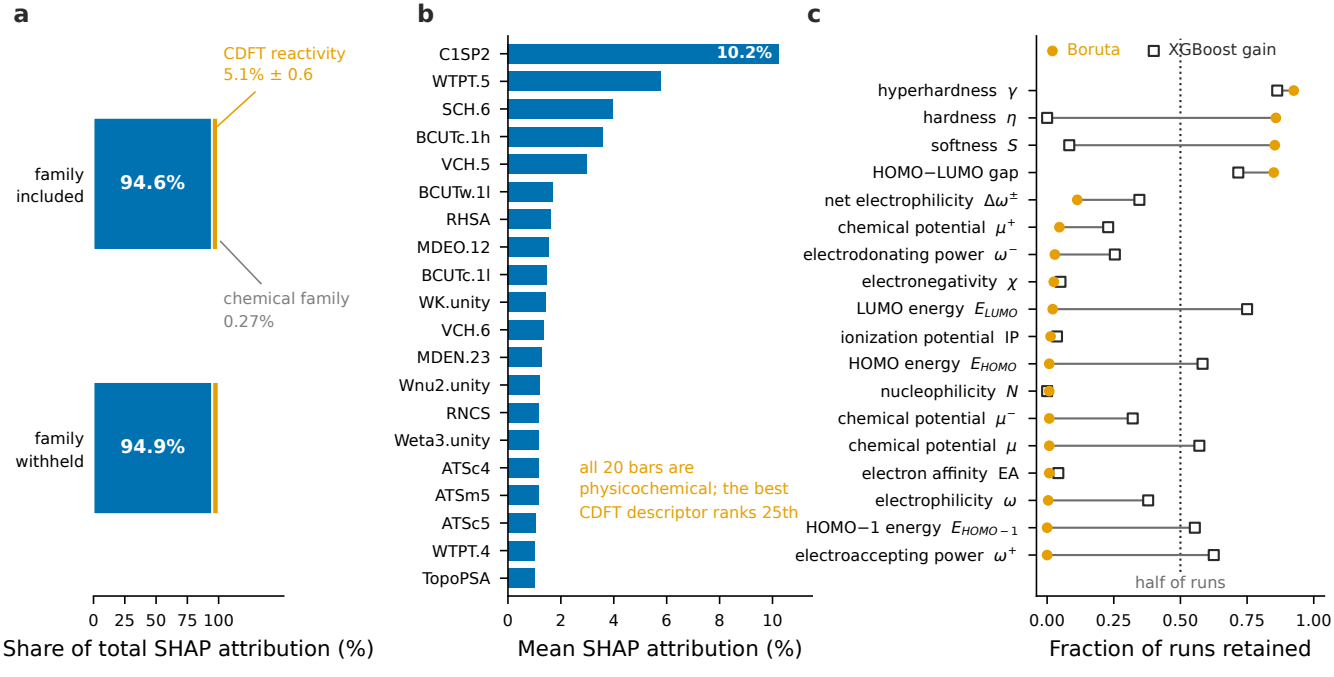
a
CDFT reactivity
5.1% ± 0.6
family
included
94.6%
chemical family
0.27%
family
withheld
94.9%
0 25 50 75 100
Share of total SHAP attribution (%)
b
C1SP2
WTPT.5
SCH.6
BCUTc.1h
VCH.5
BCUTw.1l
RHSA
MDEO.12
BCUTc.1l
WK.unity
VCH.6
MDEN.23
Wnu2.unity
RNCS
Weta3.unity
ATSc4
ATSm5
ATSc5
WTPT.4
TopoPSA
10.2%
all 20 bars are
physicochemical; the best
CDFT descriptor ranks 25th
0 2 4 6 8 10
Mean SHAP attribution (%)
c
Boruta
XGBoost gain
hyperhardness γ
hardness η
softness S
HOMO−LUMO gap
net electrophilicity Δω±
chemical potential μ+
electrodonating power ω−
electronegativity χ
LUMO energy E_LUMO
ionization potential IP
HOMO energy E_HOMO
nucleophilicity N
chemical potential μ−
chemical potential μ
electron affinity EA
electrophilicity ω
HOMO−1 energy E_HOMO−1
electroaccepting power ω+
half of runs
0.00 0.25 0.50 0.75 1.00
Fraction of runs retained